THE UNIVERSITY OF AZAD JAMMU AND KASHMIR

LAB REPORT NO. 01

SOFTWARE CONSTRUCTION & DEVELOPMENT (LAB)

COURSE TITLE	SOFTWARE CONSTRUCTION & DEVELOPMENT (LAB)
SEMESTER	Fall 2026
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DEPARTMENT	Software Engineering

DEPARTMENT OF SOFTWARE ENGINEERING

LAB 01: USE CASE DIAGRAMS

Introduction

A use case diagram is a UML (Unified Modeling Language) diagram that shows how users or external systems interact with a system. It presents the main actors, system functions, and their relationships in a simple visual form.

Objectives

- To identify actors and use cases for the given systems.
- To represent system functionality using UML use case diagrams.
- To understand how external users and devices interact with a software system.

QUESTION 1: ONLINE TICKET RESERVATION SYSTEM

1.1 Objective

To develop a use case diagram that helps describe the requirements of an online ticket reservation system.

1.2 System Overview

The online ticket reservation system allows a customer to search for tickets, check availability, reserve a ticket, make payment, and receive booking confirmation. The customer can also view or cancel an existing booking if the system supports cancellation.

1.3 Actor

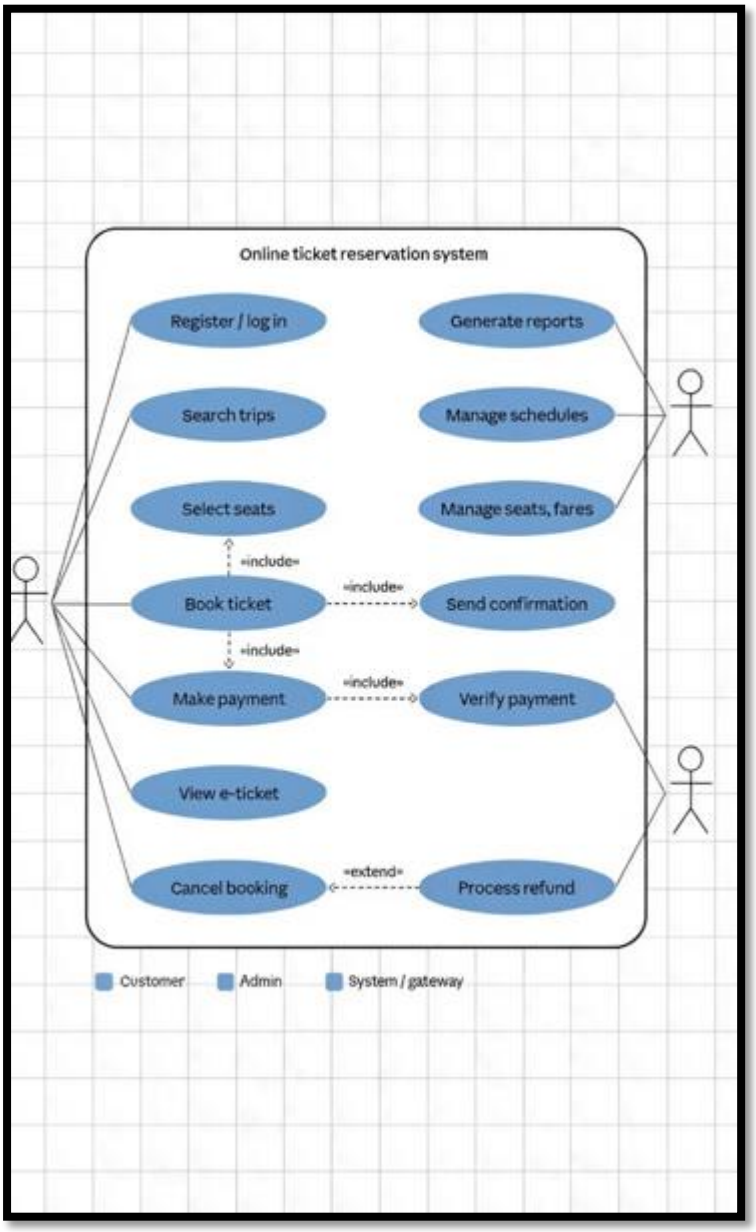
Customer: The main user who searches for tickets and performs reservation-related activities.

Payment Gateway: An external service that processes the customer's online payment.

1.4 Main Use Cases

Use Case	Description
Search Tickets	Search for tickets according to the required journey or event.
View Availability	Check whether tickets are available.
Book Ticket	Reserve a selected ticket.
Make Payment	Pay the required ticket amount.
Receive Booking Confirmation	Receive confirmation after a successful booking.
View My Booking	Review booking details.
Cancel Booking	Cancel an existing booking where cancellation is permitted.

1.5 Use Case Diagram



1.6 Explanation

The Customer is the primary actor and interacts with the reservation functions inside the system boundary. The Payment Gateway is an external actor connected to the payment function. The diagram provides a high-level view of the system; detailed rules such as refund policy, seat selection, and ticket types can be added if required.

QUESTION 2: CREDIT CARD PROCESSING SYSTEM

2.1 Objective

To identify the actors and use cases in the given credit card processing case study and represent them in a UML use case diagram.

2.2 System Overview

The customer selects an item from a shop. The vendor prepares the bill, and the customer provides a credit card for processing. The card reader processes the requested transaction and prints a receipt or statement. The vendor gives the receipt to the customer, who signs it and returns it to the vendor.

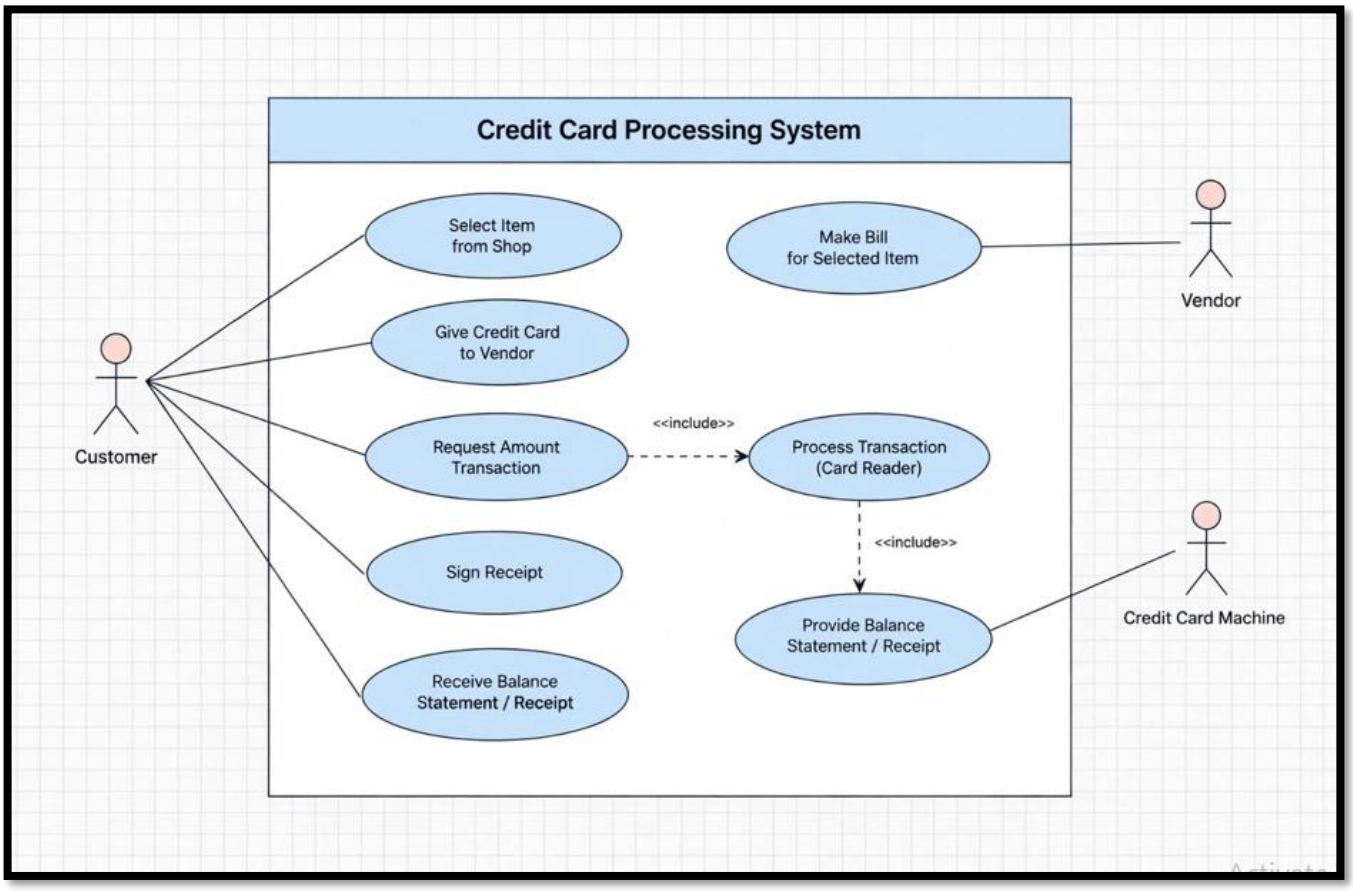
2.3 Actors

- **Customer:** Selects the item, provides the card, and signs the receipt.
- **Vendor:** Prepares the bill, operates the card reader, and gives the receipt to the customer.
- **Card Reader / Credit Card Machine:** Reads card details, processes the transaction, and prints the receipt.

2.4 Main Use Cases

Use Case	Description
Select Item	Customer chooses the item to purchase.
Generate Bill	Vendor prepares a bill for the selected item.
Swipe / Read Card	The card is presented to the card reader.
Process Transaction	The machine processes the requested payment.
Print Receipt	The machine prints the transaction receipt or statement.
Issue Receipt to Customer	Vendor hands the receipt to the customer.
Sign Receipt	Customer signs the receipt and returns it to the vendor.

2.5 Use Case Diagram



2.6 Explanation

The Customer, Vendor, and Card Reader are the actors involved in the process. The associations show which participant interacts with each function. The diagram follows the case study sequence from item selection and bill generation to payment processing, receipt printing, and signing.

Conclusion

This lab demonstrates how UML use case diagrams describe the functional requirements of software systems. The online ticket reservation diagram presents customer booking activities and payment interaction, while the credit card processing diagram represents the roles of the customer, vendor, and card reader in a retail transaction.

THANK YOU!